B Empirical validation

Here we provide details on our empirical validation of our theoretical results. All of our code was forked from the code provided in Nichani et al. [23] (the model training, plotting and logging are all nearly identical to their code; the data generating process and model itself are modified to fit our question and setting). Our code is available at `https://github.com/johnchrishays/icl/`.

B.1 Setup.

We train a disentangled transformer using the architecture described in Section 2. The only difference is that, in the main text, our results are about the (possibility or impossibility) of copying Y to the *residual stream* and to train a model, we must produce some *output*. This is to enable taking gradients and otherwise applying a standard training recipe to the inverse permutation learning problem. To generate outputs, we define output weights W and multiply the residual stream output in the last layer with W: $h^{(2)}W^\top$.

We train disentangled transformers with a causal mask and without a causal mask, as defined in Section 2.1 and Section 2.2, respectively. For each of the experiments, the training parameters, loss function and all other details of the implementation and training are exactly the same (including the random seed).

We use squared loss to optimize the model, 2^{16} training steps and batches of size 1024. Model training takes less than 5 minutes on one Nvidia A100 GPU. We set the dimension $d = 10$.

To generate the inputs, we generated P by sampling uniformly at random from the set of permutations on d elements. To generate Y, we sampled each entry of a $d \times d$ matrix uniformly at random from $\{0, 1\}$.

B.2 Results.

Transformers with causal masks. After training the model with causal mask, mean squared error of the model is approximately 2.5, which is the MSE corresponding to random guessing.

Transformers without causal masks. After training the model without the causal mask, the MSE of the model is about 0.00015. We reproduce the weights corresponding to the trained model in Figure 2. Panel (a) shows the weights for the first attention layer $A^{(1)}$, panel (b) is for the second attention layer $A^{(2)}$ and (c) is for the output layer W. The construction recovered by the model training process is not the same as that of the construction provided in our proof of Theorem 2; both are valid. We visualize weights with a heatmap: each small square in each figure represents a single weight parameter. Thus, $A^{(1)}$ is a matrix of size $3d \times 3d = 30 \times 30$. In (a), we label weights corresponding to token and position embeddings above and to the side of the matrix. Weights that are larger are closer to yellow and weights that are close to zero are dark blue. Thus, the weights matrix $A^{(1)}$ is approximately equal to

$$A^{(1)} = \beta \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & I \\ 0 & 0 & 0 \end{bmatrix}$$

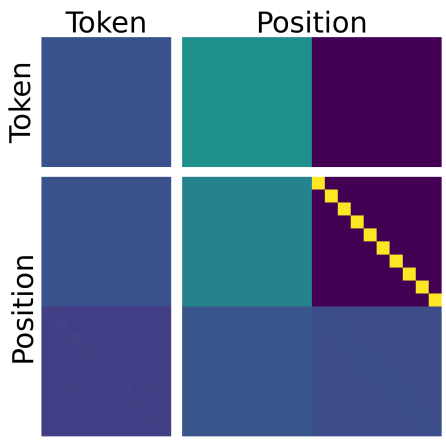
for some $\beta \gg 0$, where each entry corresponds to a $d \times d$ block.

In (b), we label weights corresponding to the different blocks of the residual stream: the top/left blocks correspond to the input sequence and the bottom/right blocks correspond to the layer-1 outputs. Thus, the weights matrix $A^{(2)}$ is approximately equal to

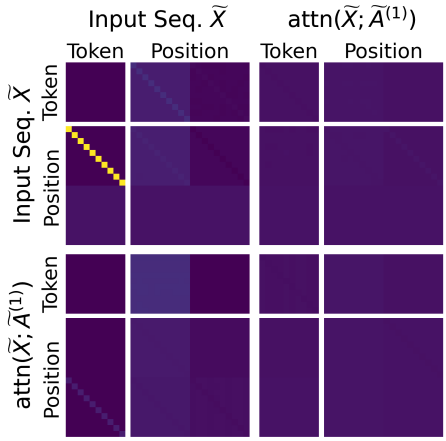
$$A^{(2)} = \beta \left[\begin{array}{ccc|ccc} 0 & 0 & 0 & 0 & 0 & 0 \\ I & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

for some $\beta \gg 0$.

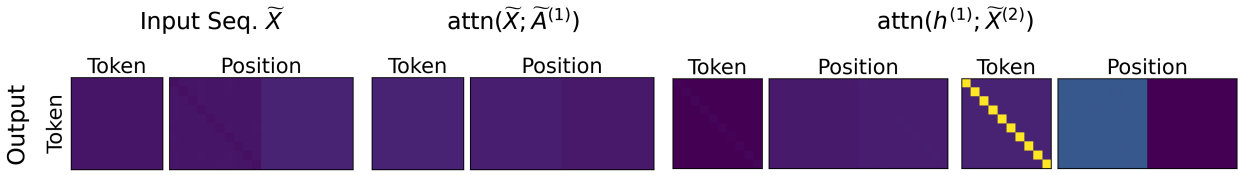
In (c), the output weights W can be seen to be approximately equal to the all-zeros matrix except for the 10th block, which is the identity matrix.



(a) Weights for first attention layer $A^{(1)}$



(b) Weights for second attention layer $A^{(2)}$

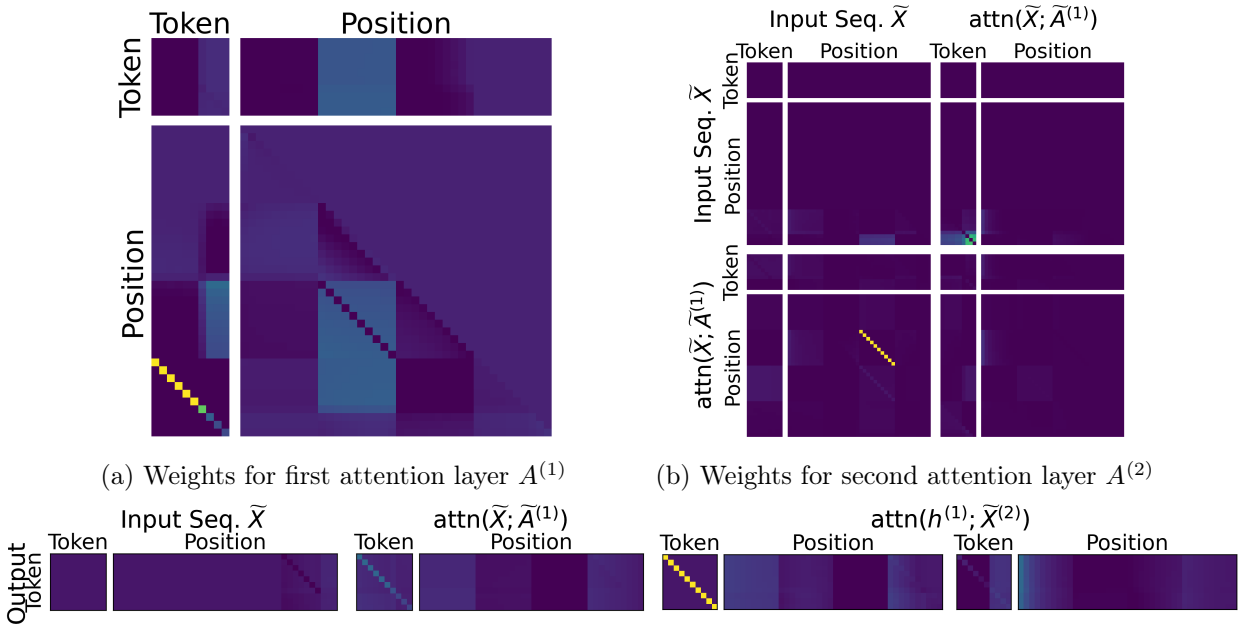


(c) Weights for output layer W

Figure 2: Weights for decoder-only causal-mask-free disentangled transformer trained to invert permutations.

These learned weights constitute a constructive proof of Theorem 2.

Transformers with scratch space. To facilitate ease of visual inspection of the trained transformers, we modify the training setup from above so that there is a d-dimensional all-zeroes matrix prepended to the input. After training the model, the test MSE is 0.00013. We reproduce the weights corresponding to the trained model in Figure 3. The panels display the same quantities as in the case of the causal-mask-free transformer. These weights constitute an alternative constructive proof of Theorem 3.



(a) Weights for first attention layer $A^{(1)}$

(b) Weights for second attention layer $A^{(2)}$

(c) Weights for output layer W

Figure 3: Weights for decoder-only disentangled transformer with scratch space trained to invert permutations.

C Proofs

We begin by stating a lemma we will use in the proof of our main theorem. It formalizes the notion that the causal mask induces an invariance of each row to any subsequent row in the residual stream.

Lemma 1. *Let h and h' be any two inputs to the ℓth layer such that their first i rows are the same: $h_{1:i} = h'_{1:i}$. Then for any weights A, $\mathrm{attn}(h; A)_i = \mathrm{attn}(h'; A)_i$.*

Proof. By inspecting the activation function, we observe the ith row of the output of a given layer with weights A are:

$$(\mathcal{S}(\mathrm{MASK}(hAh^\top))h)_i = \frac{1}{\sum_{j \le i} \exp(h_i A h_j^\top)} \sum_{j \le i} \exp(h_i A h_j^\top) h_j$$

Since this function does not depend on any row h_j for $j > i$, $\mathrm{attn}(h; A)_\ell = \mathrm{attn}(h'; A)_\ell$. □

C.1 Proof of Theorem 1

For ease of reference, we restate each result before its proof.

Theorem 1. *For all k and parameter matrices $\{A^{(i)}\}_{i \in [k]}$, and all permutation matrices P other than the identity permutation, there exists a target matrix Y such that a decoder-only, attention-only transformer parameterized by $\{A^{(i)}\}_{i \in [k]}$ given $[P; Y_P]$ as input does not output Y to any block of the residual stream.*